

Assessment Test: Software Tester (QA)

Task Management Application — Comprehensive Test Suite

Part 1: Test Case Design

1. User Registration & Login

Scenario: Verify user registration with valid and invalid data.

- **Positive:** Successfully register with a valid email, unique username, and strong password.
- **Negative:** Attempt registration leaving mandatory fields (Email, Password) blank. Verify specific error messages.
- **Edge Case:** Register with an email address containing unusual but valid characters (e.g., user.name+tag@domain.co.uk).
- **Edge Case:** Attempt to register an already existing email address. Verify that the system blocks it and displays a 'User already exists' warning.

Scenario: Verify user authentication and session creation.

- **Positive:** Log in successfully using valid registered credentials and ensure redirection to the task dashboard.
- **Negative:** Attempt login with an incorrect password or an unregistered email. Ensure access is denied.
- **Security:** Enter SQL injection strings (e.g., ' OR '1'='1) into the login fields. Ensure the application handles input securely without logging the user in.
- **Edge Case:** Test the application's behavior when a user attempts to log in multiple times simultaneously from different browser tabs or devices.

2. Task CRUD Operations

Scenario: Verify task creation under various input constraints.

- **Positive:** Create a task with a valid title, description, and future due date. Verify it saves successfully to the database.
- **Negative:** Attempt to save a task with an empty title field. Verify that saving is blocked.
- **Edge Case:** Enter extremely long text (e.g., 5000+ characters) into the task title and description fields to check for character limit handling and UI distortion.

Scenario: Verify the accuracy, loading performance, and rendering of the task list.

- **Positive:** Verify that all created tasks populate accurately in the list with the correct details matching the database records.
- **Edge Case:** Test performance and pagination UI when a user has a massive volume of tasks (e.g., 10,000+ tasks).

Scenario: Verify task modification capabilities.

- **Positive:** Modify the title and status of an existing task, save it, and verify the changes persist after refreshing the page.

- **Edge Case:** Open a task to edit in two separate browser windows simultaneously, save edits in both, and verify how the system handles the data conflict (Concurrency test).

Scenario: Verify task removal and data integrity.

- **Positive:** Delete a task and confirm a confirmation modal pops up. Upon confirming, ensure the item is permanently removed from the UI list and the database.
- **Negative:** Cancel the deletion mid-process via the confirmation modal. Ensure the task remains unaltered in the list.

Part 2: Bug Identification & Risk Areas

Bug Description	Severity	Reason / Impact
Lack of validation against past dates during task creation.	Major	Users could accidentally set task deadlines in the past (e.g., yesterday). This breaks the core business logic of scheduling and notifications.
Failure to restrict task CRUD operations across different user accounts via API manipulation.	Critical	An authenticated user might alter URL parameters or API requests (e.g., from id 10 to 11) to access or delete tasks belonging to another user (IDOR vulnerability).
Database text truncation error or UI crash when saving special characters/emojis.	Minor	If the database collation does not support full UTF-8 encoding (like emojis), saving a task containing emojis could throw an unhandled 500 error or crash the task dashboard UI.
Multi-click form submission leading to duplicate entries.	Major	Rapidly clicking the 'Save Task' button multiple times before the server completes processing could create duplicate identical tasks in the database, cluttering user data.